

The Invisible Labor Index

Measuring Emotional Work in Public Service Jobs — Summary Report

Executive Summary

This report presents findings from the Invisible Labor Index (ILI), a composite measure of emotional and psychological labor derived from **1,543** Glassdoor reviews across **5 occupations** (Nurse, Teacher, Driver, Sales Associate, Social services worker) and **4 U.S. regions** (West, Northeast, South, Midwest). The index was built using zero-shot text classification to score each review on nine emotional-labor constructs, then combined into two composite scores: an Invisible Labor Score (burnout, compassion fatigue, management frustration, pay dissatisfaction, turnover) and a Positive Experience Score (job satisfaction, team support, pride, work-life balance).

Across the full sample, the average Invisible Labor Score was **0.45** and the average Positive Experience Score was **1.07** — indicating that, on balance, positive emotional signals (fulfillment, support, satisfaction) outweighed strain signals (burnout, frustration) across reviews. The average star rating was **3.37** out of 5.

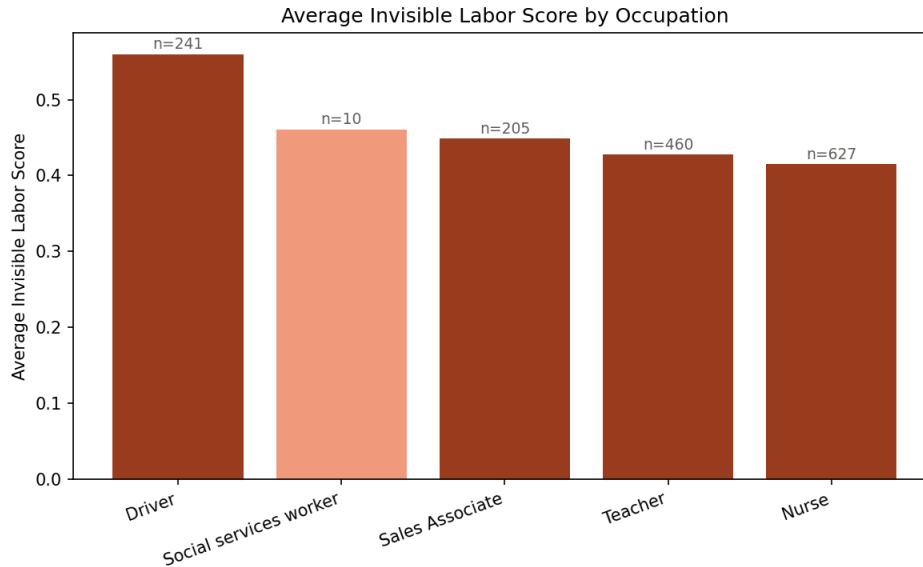
Headline finding: Drivers show the highest average Invisible Labor Score (0.560) of any occupation studied — higher than Nurses (0.415), which in fact scored lowest. This runs counter to the common assumption that healthcare work carries the heaviest emotional burden; in this dataset, driving-related roles showed comparatively more burnout and dissatisfaction signal in review text.

Methodology

Review text (pros/cons) from Glassdoor was classified using a zero-shot NLP model (facebook/bart-large-mnli) against nine emotional-labor constructs, producing a continuous probability score (0–1) per construct per review. These were combined into weighted composite scores. Salary data was joined at the occupation level via keyword-based title mapping; demographic ratings were joined at the company level from Glassdoor's aggregate demographic breakdown.

Key Findings

1. Emotional labor varies significantly by occupation



Occupation	Avg ILI	Avg Positive	Avg Rating	n
Driver	0.560	0.973	3.033	241
Social services worker*	0.461	0.359	3.340	10
Sales Associate	0.449	1.002	3.200	205
Teacher	0.428	1.031	3.419	460
Nurse	0.415	1.172	3.522	627

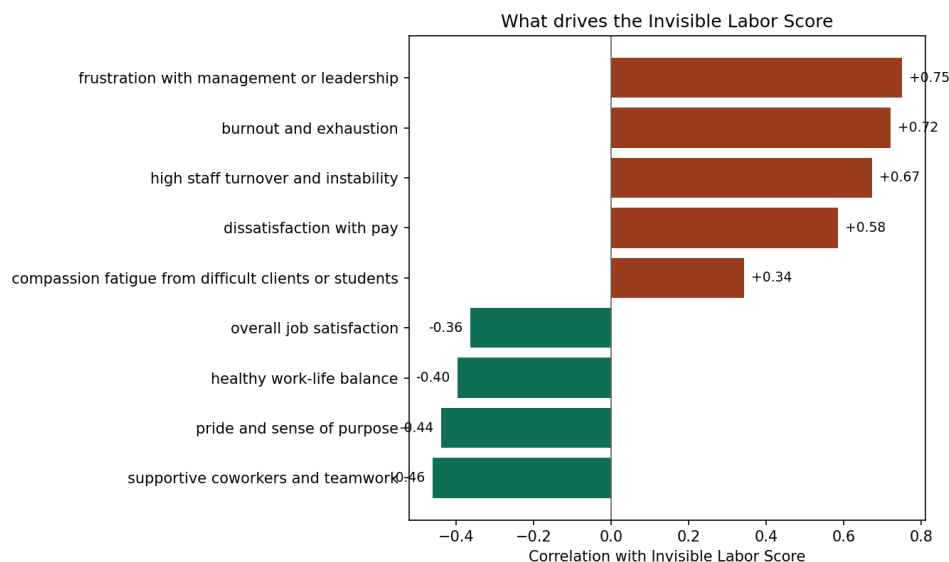
**Social services worker has only 10 reviews — too small a sample to draw a reliable conclusion from; shown for completeness but excluded from headline claims.*

- Highest average Invisible Labor Score: **Driver (0.560, n=241)**
- Lowest average Invisible Labor Score: **Nurse (0.415, n=627)**
- Highest average Positive Experience Score: **Nurse (1.172, n=627)**

Notably, Nurse scores lowest on the Invisible Labor Score and highest on the Positive Experience Score simultaneously — despite having a large sample (n=627), suggesting nursing reviews skew toward describing the work as demanding but rewarding, rather than purely burdensome.

2. What drives the Invisible Labor Score

Correlation analysis across all nine emotional-labor constructs identified three dominant drivers of a high Invisible Labor Score:



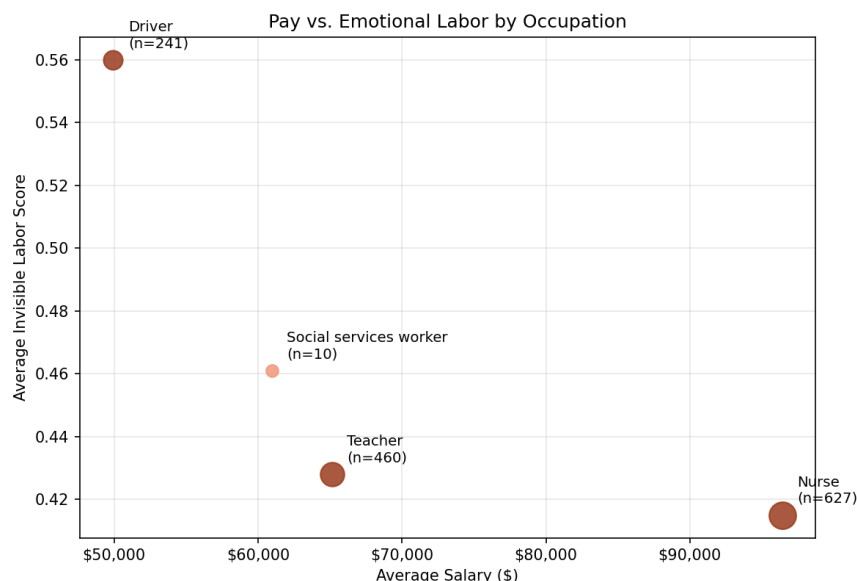
- Frustration with management or leadership (**$r = 0.75$**)
- Burnout and exhaustion (**$r = 0.72$**)
- High staff turnover and instability (**$r = 0.67$**)

Pay dissatisfaction followed closely ($r = 0.59$). On the opposite end, supportive coworkers ($r = -0.46$), pride and sense of purpose ($r = -0.44$), and healthy work-life balance ($r = -0.40$) were most strongly associated with a *lower* Invisible Labor Score.

Notable secondary finding: star rating correlates with the Invisible Labor Score at only $r = -0.53$, and salary correlates at just $r = -0.03$ — meaning traditional metrics (star ratings, pay) capture only part of the emotional labor picture. This is direct evidence for this project's core premise: emotional labor is genuinely under-measured by conventional workplace metrics.

3. Pay and emotional labor

Individual review-level salary showed almost no correlation with the Invisible Labor Score ($r = -0.03$) — but at the *occupation* level, a clear pattern emerges:

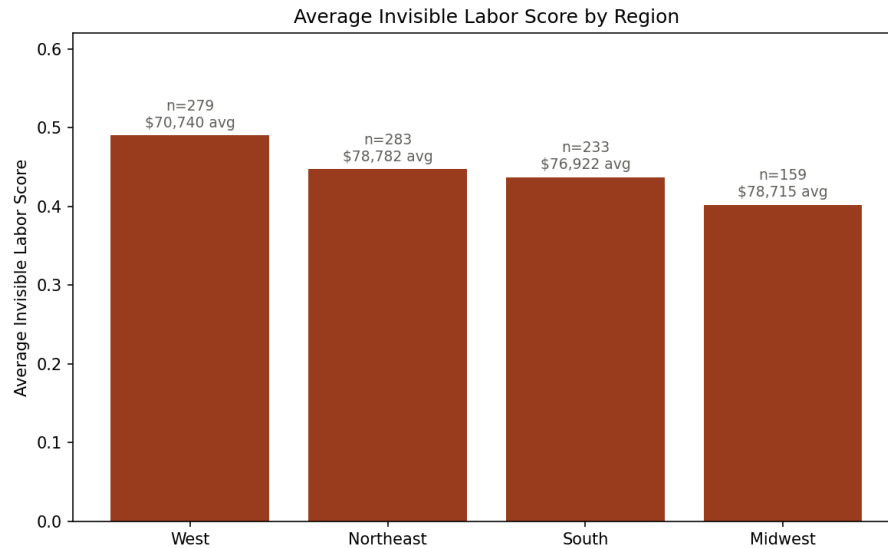


- Driver is both the **lowest-paid** occupation studied (\$49,927 average) and has the **highest** Invisible Labor Score (0.560).
- Nurse is the **highest-paid** occupation (\$96,406 average) and has one of the **lowest** Invisible Labor Scores (0.415).
- Sales Associate had no matched salary data in this dataset (a data-collection gap, not a zero) and is excluded from this comparison.

The apparent contradiction — no relationship at the review level, but a clear relationship at the occupation level — is explained by the fact that pay is fairly uniform *within* an occupation/company, so individual-review variation in salary is small relative to variation across occupations. The occupation-level pattern is the more meaningful signal here: it suggests that emotional labor is not being compensated proportionally to its intensity — the occupation with the most emotionally taxing reviews is also the lowest paid.

4. Regional patterns

Region was derived from each review's state (mapped to the standard four U.S. Census regions). **Note: 38.2% of reviews (589 of 1,543) had no region assigned**, almost entirely because location data was missing from the source review (580 of the 589 had no state recorded at all) rather than a mapping error. Regional findings below are based on the 954 reviews with a known region.



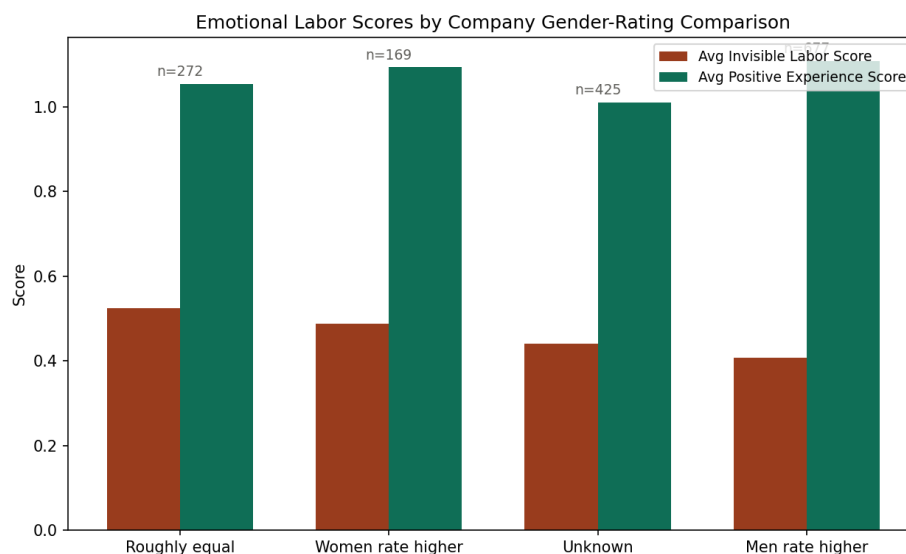
- Highest Invisible Labor Score: **West (0.490, n=279)** — despite the West also having the lowest average salary of the four regions (\$70,740).
- Lowest Invisible Labor Score: **Midwest (0.402, n=159)**

The regional pattern echoes the occupational one: the region with the lowest average pay (West) also shows the highest emotional labor score, while better-paid regions (Midwest, Northeast) trend lower — reinforcing the pay-equity signal found at the occupation level, though the regional gap is narrower than the occupational one.

Workforce Equity Implications

Company-level demographic ratings were used to group companies by whether women or men rate that employer higher on Glassdoor overall, then compared against review-derived emotional labor scores.

27.5% of reviews (425 of 1,543) fall into an "Unknown" group because the source company lacked a men/women rating breakdown on Glassdoor.



- Companies where **men rate the employer higher** than women show the **lowest** average Invisible Labor Score (0.407, n=677 — the largest group).
- Companies where **women rate the employer higher** show a moderate score (0.488, n=169).
- Companies where ratings are **roughly equal** between genders show the **highest** score (0.525, n=272).

Interpretation: this result does not support a simple linear story (e.g. "lower relative ratings for women predict higher emotional labor"). The relationship is non-monotonic — the highest Invisible Labor Score occurs in companies where men and women rate the employer roughly equally, not where the gap favors one gender. This may reflect a genuinely more complex relationship, or may simply reflect the coarseness of a company-level (not individual-level) gender signal — see Limitations. Positive Experience Scores, by contrast, were relatively stable across all four groups (1.01–1.11), suggesting the gender-rating gap relates more to strain-side experiences than to positive ones. This finding is presented as-is rather than forced into a cleaner narrative than the data supports.

Limitations

- **Company-level, not individual-level, demographics:** Gender, race, and other demographic ratings come from Glassdoor's aggregate company breakdown, not from the individual reviewer. Every review from the same company shares identical demographic values, so equity findings are directional, not precise per-person measurements.
- **Classification is model-based, not ground truth:** Zero-shot classification scores are the model's confidence that a construct applies to the text, not a verified human judgment. Scores were not independently validated against human-labeled data.
- **Glassdoor selection bias:** Reviewers self-select into leaving reviews, typically skewing toward strongly positive or strongly negative experiences rather than a representative sample of all employees.
- **Salary-to-occupation mapping:** Salary figures were matched to occupations via keyword matching on free-text job titles, which may misclassify some roles.
- **Small-sample instability:** Some occupation/region/company combinations have few reviews; scores for these groups should be treated with lower confidence than high-volume groups.
- **Composite score weights are researcher-chosen:** The relative weighting of each construct in the Invisible Labor Score reflects a reasoned but subjective judgment, not an empirically derived optimum.

Conclusion

The Invisible Labor Index reveals that emotional labor is real, measurable, and unevenly distributed — but not always where intuition expects it. Driver, not Nurse or Teacher, carries the heaviest emotional burden by this measure, and that burden tracks closely with pay: the lowest-paid occupation and region in this dataset are also the most emotionally taxed. Conventional metrics — star ratings, salary — capture only part of this picture ($r = -0.53$ and $r = -0.03$ respectively), which is precisely the gap this project set out to close. The gender-equity findings are less conclusive and should be extended with individual-level data before drawing policy conclusions. Future work — incorporating BLS occupational statistics, individual-level demographic data, and a larger, less Glassdoor-selection-biased sample — would meaningfully sharpen these findings for employers and policymakers focused on compensating emotional labor fairly.